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RIVER VALLEY HIGH SCHOOL
General Certificate of Education Advanced Level
Preliminary Examination 2024
Higher 2

GEOGRAPHY 9173/02

Paper 2 **12 September 2024**

3 hours

Additional Materials: 1 Insert

READ THESE INSTRUCTIONS FIRST

An answer booklet will be provided with this question paper. You should follow the instructions on the front cover of the answer booklet. If you need additional answer paper ask the invigilator for a continuation booklet.

The Insert contains all the Resources referred to in the questions.
You should make reference to appropriate examples studied in the field or the classroom, even where such examples are not specifically requested by the question.
Diagrams and sketch maps should be drawn whenever they serve to illustrate an answer.
You are reminded of the need for good English and clear presentation in your answers.

The number of marks is given in brackets [ ] at the end of each question or part question.

Section A

Answer the question found in this section.

Cluster 4: Fieldwork

1 As part of a class project investigating the needs of the elderly in Bukit Merah estate, a group of eight junior college students chose to investigate the effectiveness of a "Silver Zone" retrofit of a stretch of road at Bukit Merah View, a housing estate with a prominent elderly demographic profile.

Silver Zones are areas designated by the Land Transport Authority (LTA) with enhanced road safety measures which make it safer and more convenient for senior pedestrians to cross the roads. These features include distinctive signs, road features and markings which help to lower vehicle speeds and guide pedestrians to designated crossing points.

The group decided to investigate the effectiveness of the Silver Zone features in Bukit Merah View.

They crafted a questionnaire survey asking their respondents on their current perceptions of their neighbourhood's accessibility as well as their opinions on the differences made by the Silver Zone features.

Their data collection plan involved them breaking into two groups of four students, groups A and B. Group A would gather responses through convenience sampling at the void decks of apartment blocks adjacent to the Silver Zone, while Group B would ply the pedestrian walkways along the Silver Zone to gather responses through convenience sampling. In accommodating their school timetable, the group planned to carry out their fieldwork over three weekday afternoons between 3pm and 7pm.

On the third session of their fieldwork, stormy weather forced Group B to take shelter, causing them to start only after the weather has cleared at 6pm.

Resource 1 shows a map of the estate and the sites of data collection in the group's plan.
Resource 2 outlines the key Silver Zone features implemented at the site.
Resource 3 shows the questionnaire survey drafted by the students.
Resource 4 shows raw data of the students' findings.

(a) Explain a suitable hypothesis for the students' fieldwork investigation. [4]

(b) Explain **three** safety considerations the students should have planned for in their fieldwork. [6]

(c) Citing data from Resource 1 and the preamble, explain how the students' data collection method could be improved. [6]

(d) Citing information from Resource 3, explain the strengths and limitations of the questionnaire the group designed. [6]

(e) Using a suitable form of data representation, present the group's findings as shown in Resource 4. [4]

(f) Citing data from Resource 4, compare the findings from Groups A and B. [4]

(g) Evaluate the usefulness of the group's fieldwork findings to the class project on understanding the needs of the elderly in Bukit Merah estate. [10]

Section B

Cluster 3: Sustainable Future and Climate Change

2 Resource 5 shows the global temperatures since 1880 as compared to the average global temperature for the mid-20th century years of 1951-1980. Resource 6 shows a simplified schematic representation of the greenhouse effect. Resource 7 shows information about the ecological footprint, Human Development Index (HDI) and Happy Planet Index in selected countries for 2016. Resource 8 is a table showing information about the ecological footprint, population and GNI per person data for selected groups of countries.

(a) With reference to Resource 5, explain the possible effects of global temperature changes on aquatic and terrestrial ecosystems. [8]

(b) Explain the limitations of Resource 6 as a source of information about the greenhouse effect. [3]

(c) Explain how human activities have accelerated global temperature changes through positive feedbacks. [6]

(d) Cite data from Resource 7 to describe the relationships shown. [5]

(e) Cite data from Resource 8 to describe the variations in ecological footprints between the groups of countries shown. [4]

(f) With reference to Resources 7 and 8, explain why the ecological footprint differs across countries and regions. [4]

Section C

Answer **either** question 3 **or** question 4.

Cluster 3: Sustainable Future and Climate Change

3 Evaluate the impact of urban reimaging on sustainable urban development. [20]

4 'Uncertainties over future impacts of climate change are the greatest impediment to the implementation of effective strategies to respond to climate change.' Evaluate this statement. [20]

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